

CHRISANTUS EZE

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EDUCATION

Oklahoma State University (OSU) | Ph.D., Computer Science (in-view) *January 2022 - Oct 2026*

- Research: World models, VLA/foundation models, self-supervised representation learning, formal reasoning (Answer Set Programming), LoRA fine-tuning, ROS, MuJoCo, PyBullet, PyTorch, HuggingFace, LangChain, AWS, Docker, Python

Federal University of Technology, Owerri (FUTO), Nigeria | B.Eng. in Electrical & Electronic Eng. *October 2013 - October 2018*

RELEVANT EXPERIENCE

Graduate Researcher – Machine Learning & AI | Department of Computer Science, OSU *January 2022 - Present*

- **World models for embodied prediction and control:** designing and training self-supervised latent-space world models and dynamics models from scratch, learning predictive representations over visual and proprioceptive input to support long-horizon planning and control for vision-language-action (VLA) policies.
- Lead research on multimodal foundation models for long-horizon spatial reasoning and robotic manipulation.
- **Sim-to-real manipulation, physics-simulator-validated:** developed Unveiler, a planning-control framework for sequential object retrieval in clutter, achieving 94% sim-to-real transfer success under real-world perceptual noise.
- **Formal reasoning for verification:** developing logic-augmented spatial reasoning for VLMs via Answer Set Programming (ASP) — a formal logic-based knowledge representation and reasoning paradigm — benchmarking zero-shot performance and fine-tuning on curated spatial datasets to improve compositional and relational reasoning.
- **Data curation and grounding:** built VLM-RAG pipelines in PyTorch for object grounding (+34% improvement); developed a source-free domain adaptation method (A3) enabling policy transfer to novel environments without target-domain supervision (+15% cross-domain performance).
- **Foundation-model fine-tuning:** deployed and fine-tuned instruction-tuned VLMs using LoRA with PEFT, gradient checkpointing, and mixed-precision training, enabling GPU-efficient experimentation and private model adaptation.
- Published 5 peer-reviewed papers in ML/robotics venues.

Engineering Intern – Applied ML for Process Systems | Fractionation Research Inc. *June 2025 - August 2025*

- Architected a real-time anomaly detection system for distributed control (50+ sensor streams), reducing false positives by 78% and contributing to a 23% reduction in unplanned downtime.

Graduate Student Mentee – Applied ML Research | Google *February 2023 - June 2025*

- Co-developed sequential manipulation policies using visual reasoning, behavior cloning, and reinforcement learning (98% benchmark success).
- Contributed a perception module integrating depth-based occlusion reasoning and learned affordances, part of a multi-modal sensor fusion pipeline for manipulation.

RELEVANT PUBLICATIONS

- Eze, C. and Crick, C. "Verify2Act: Critic-Guided Latent World Model For Scalable Long Horizon Planning." [Under review]
- Eze, C., Julian, R., and Crick, C. "Unveiler: Modular Spatial Reasoning for Sequential Manipulation in Dense Clutter." IROS 2026
- Eze, C. and Crick, C. "Learning by Watching: A Review of Video-based Learning Approaches for Robot Manipulation." IEEE Access, 2025.
- Ezeakunne, U., Eze, C., and Liu, X. "Data-Driven Fairness Generalization for Deepfake Detection." ICAART 2025.
- Eze, C. and Crick, C. "A3: Active Adversarial Alignment for Source-Free Domain Adaptation." IEEE ICMLA 2024.
- Eze, C. and Crick, C. "Enhancing Human-Robot Collaboration by Exploring Intuitive Augmented Reality Design Representations." ACM/IEEE HRI 2023.

NON-RESEARCH PROJECTS

Deep-Research Companion

- Built an agentic research collaborator that takes a hypothesis, retrieves literature from arXiv and Semantic Scholar, builds an evidence-linked belief state, and pressure-tests it through Skeptic/Optimist/Engineer roles, iterating via a reflexion loop, all running locally via Ollama. Exposed the pipeline as an MCP server with stateful, multi-turn tools.

LLM-Powered Research Assistant (Lab-Mate)

- Built a GPT-4o Slack bot to auto-summarize and extract action items from over 100 research meetings.
- Built a RAG pipeline using semantic chunking and vector search, boosting contextual retrieval of lab protocols and discussions to 89% relevance; end-to-end pipeline ownership (curation, embedding, retrieval) transferable to sensor and trajectory data curation.